EXPERIMENT 2

Title: Implementation of a Finite State Automaton

Aim:

To implement a Finite State Automaton (FSA) using Python to recognize strings that end with the sequence ab. A finite state automaton is a computational model that processes an input string character by character and determines whether the string satisfies a particular pattern. It is commonly used in lexical analysis, pattern recognition, and language processing.

Algorithm:

1. Start the program.
2. Define a function named ends_with_ab() that accepts a string as input.
3. Initialize the automaton in the starting state q0.
4. Read each character of the input string one by one.
5. Change the current state according to the character being processed.
6. Use q1 to represent that the latest relevant character is a.
7. Use q2 to represent that the string currently ends with ab.
8. If additional characters are encountered, update the state according to the transition rules.
9. After processing the complete string, check whether the final state is q2.
10. If the final state is q2, accept the string; otherwise, reject it.
11. Test the function using different input strings and display the results.
12. Stop the program.

Program:

Exp-02-Basic Automaton.py - C:\Users\jerin\OneDrive\Desktop\NLP\Experiments\Exp-02-Ba...

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```
def ends_with_ab(text):
    state = "q0"

    for ch in text:
        if state == "q0":
            state = "q1" if ch == "a" else "q0"

        elif state == "q1":
            if ch == "a":
                state = "q1"
            elif ch == "b":
                state = "q2"
            else:
                state = "q0"

        else:
            state = "q1" if ch == "a" else "q0"

    return state == "q2"


for s in ["ab", "aab", "abc", "abab", "cab"]:
    print(s, "->",
          "Accepted" if ends_with_ab(s) else "Rejected")
```

Output:

```
>>>
    = RESTART: C:\Users\jerin\OneDrive\Desktop\NLP\Experiments\
    n.py
    ab -> Accepted
    aab -> Accepted
    abc -> Rejected
    abab -> Accepted
    cab -> Accepted
>>>
```

Result:

Thus, a Finite State Automaton was implemented successfully using Python to recognize strings ending with ab. The program processes each character using state transitions and determines acceptance based on the final state. This demonstrates the use of finite-state models for recognizing simple patterns in language processing.